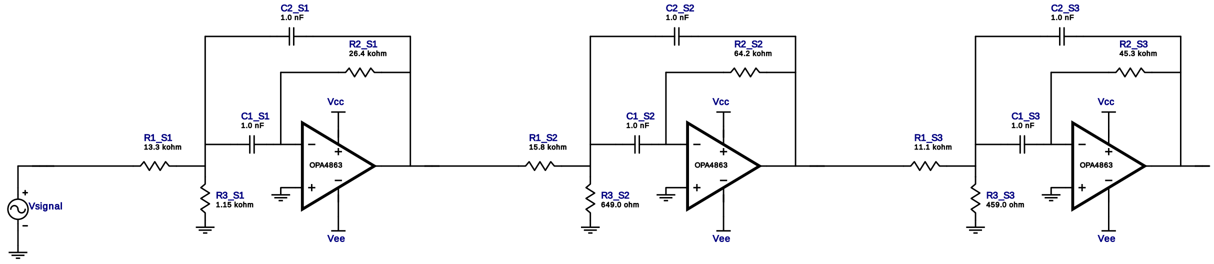
Type : Bandpass
Response : Butterworth
Order : 6
Number of Stages : 3

Filter Design Report

Design : Bandpass Filter - 6th order Butterworth
Design ID: 10



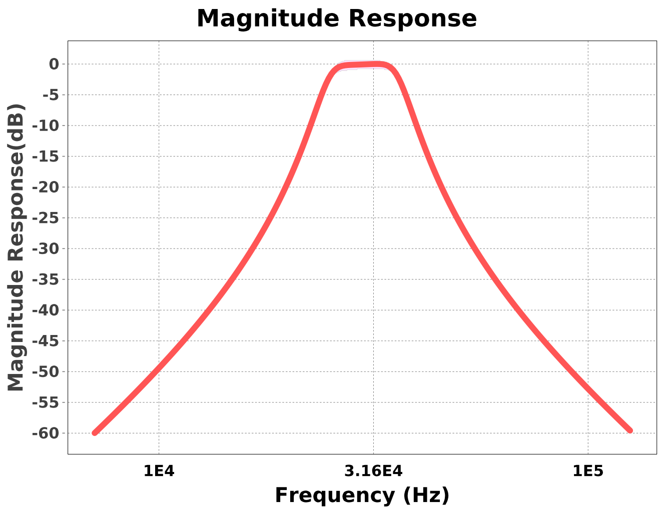
Electrical BOM

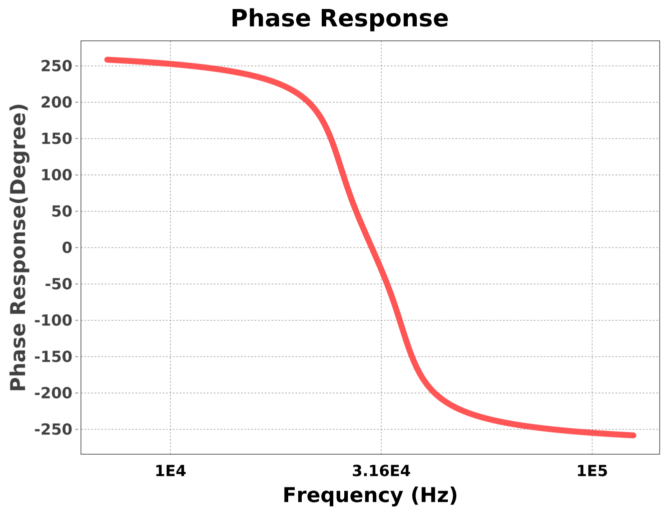
#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	OPA4863	GbwTyp= 50MHz VccMax= 12.6V VccMin= 2.7V	1
2.	A1_S2	Texas Instruments Inc.	OPA4863	GbwTyp= 50MHz VccMax= 12.6V VccMin= 2.7V	1
3.	A1_S3	Texas Instruments Inc.	OPA4863	GbwTyp= 50MHz VccMax= 12.6V VccMin= 2.7V	1
4.	C1_S1	Generic	Ideal	Cap= 1.0 nF Tolerance= 1.0 %	1
5.	C1_S2	Generic	Ideal	Cap= 1.0 nF Tolerance= 1.0 %	1
6.	C1_S3	Generic	Ideal	Cap= 1.0 nF Tolerance= 1.0 %	1
7.	C2_S1	Generic	Ideal	Cap= 1.0 nF Tolerance= 1.0 %	1
8.	C2_S2	Generic	Ideal	Cap= 1.0 nF Tolerance= 1.0 %	1
9.	C2_S3	Generic	Ideal	Cap= 1.0 nF Tolerance= 1.0 %	1
10.	R1_S1	Generic	Ideal	Res= 13300.0ohm Tolerance= 0.5%	1
11.	R1_S2	Generic	Ideal	Res= 15800.0ohm Tolerance= 0.5%	1
12.	R1_S3	Generic	Ideal	Res= 11100.0ohm Tolerance= 0.5%	1
13.	R2_S1	Generic	Ideal	Res= 26400.0ohm Tolerance= 0.5%	1
14.	R2_S2	Generic	Ideal	Res= 64200.0ohm Tolerance= 0.5%	1
15.	R2_S3	Generic	Ideal	Res= 45300.0ohm Tolerance= 0.5%	1
16.	R3_S1	Generic	Ideal	Res= 1150.0ohm Tolerance= 0.5%	1

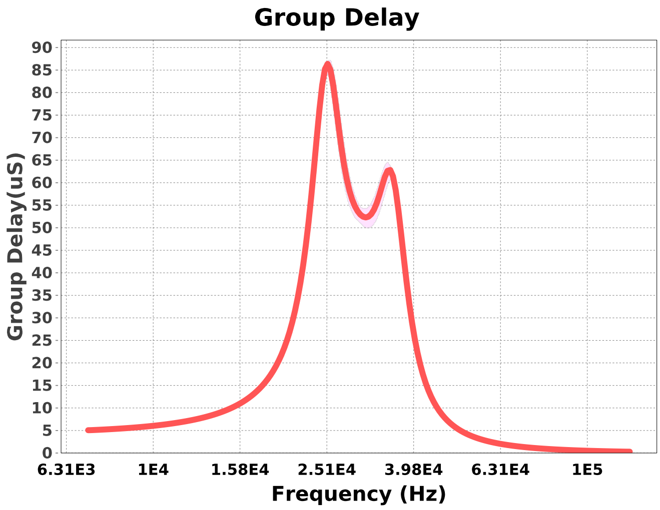
#	Name	Manufacturer	Part Number	Properties	Qty
17.	R3_S2	Generic	Ideal	Res= 649.0ohm Tolerance= 0.5%	1
18.	R3_S3	Generic	Ideal	Res= 459.0ohm Tolerance= 0.5%	1

Sensitivity Analysis

#	Name	Series	Tolerance
1.	Cap	E48	1%
2.	Res	E192	0.5%







Design Inputs

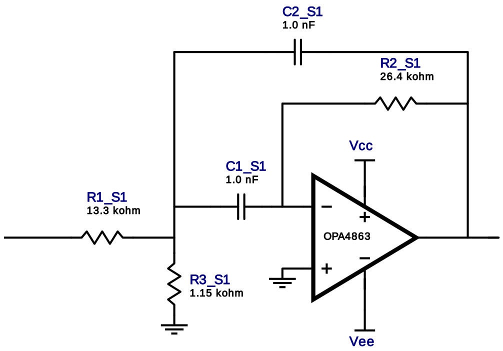
#	Name	Value	Description
1.	FilterType	bandpass	
2.	FilterResponse	Butterworth	
3.	FilterOrder	6.0	
4.	FilterTopology	Multiple Feedback	
5.	NumberOfStages	3.0	
6.	CenterFrequency	30.0 k	
7.	StopbandAttenuation	-60.0	
8.	PassbandBandwidth	12.0 k	
9.	StopbandBandwidth	120.0 k	
10.	Gain	1.0	
11.	DualSupply	+/-3.30 V	Power supply(s) to active chips
12.	ResistorTolerance	E192	Resistor series - 0.5% Passive resistor tolerance
13.	CapacitorTolerance	E48	Capacitor series - 1% Passive capacitor tolerance

Design Assistance

1. **OPA4863** Product Folder : http://www.ti.com/product/OPA4863 : contains the data sheet and other resources.

Filter Stage :1

Cutoff Frequency	30.108 kHz
Min GBW Reqd	7.5 MHz
Stage Gain	992.481 mV/V
Stage Q	2.497
Stage Topology	Multiple Feedback

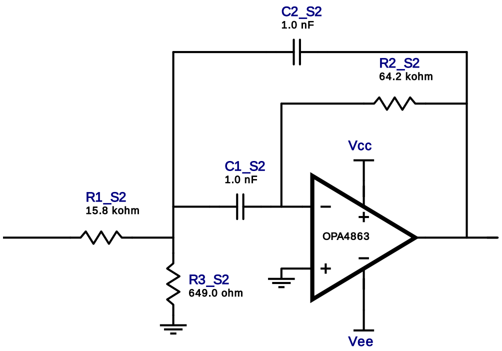


Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	OPA4863	GbwTyp= 50MHz VccMax= 12.6V VccMin= 2.7V	1
2.	C1_S1	Generic	Ideal	Cap= 1.0 nF Tolerance= 1.0 %	1
3.	C2_S1	Generic	Ideal	Cap= 1.0 nF Tolerance= 1.0 %	1
4.	R1_S1	Generic	Ideal	Res= 13300.0ohm Tolerance= 0.5%	1
5.	R2_S1	Generic	Ideal	Res= 26400.0ohm Tolerance= 0.5%	1
6.	R3_S1	Generic	Ideal	Res= 1150.0ohm Tolerance= 0.5%	1

Filter Stage :2

Cutoff Frequency	25.158 kHz
Min GBW Reqd	25.993 MHz
Stage Gain	2.032 V/V
Stage Q	5.074
Stage Topology	Multiple Feedback

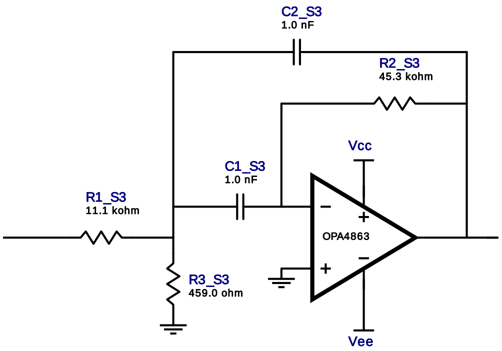


Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S2	Texas Instruments Inc.	OPA4863	GbwTyp= 50MHz VccMax= 12.6V VccMin= 2.7V	1
2.	C1_S2	Generic	Ideal	Cap= 1.0 nF Tolerance= 1.0 %	1
3.	C2_S2	Generic	Ideal	Cap= 1.0 nF Tolerance= 1.0 %	1
4.	R1_S2	Generic	Ideal	Res= 15800.0ohm Tolerance= 0.5%	1
5.	R2_S2	Generic	Ideal	Res= 64200.0ohm Tolerance= 0.5%	1
6.	R3_S2	Generic	Ideal	Res= 649.0ohm Tolerance= 0.5%	1

Filter Stage :3

Cutoff Frequency	35.617 kHz
Min GBW Reqd	36.752 MHz
Stage Gain	2.041 V/V
Stage Q	5.069
Stage Topology	Multiple Feedback



Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S3	Texas Instruments Inc.	OPA4863	GbwTyp= 50MHz VccMax= 12.6V VccMin= 2.7V	1
2.	C1_S3	Generic	Ideal	Cap= 1.0 nF Tolerance= 1.0 %	1
3.	C2_S3	Generic	Ideal	Cap= 1.0 nF Tolerance= 1.0 %	1
4.	R1_S3	Generic	Ideal	Res= 11100.0ohm Tolerance= 0.5%	1
5.	R2_S3	Generic	Ideal	Res= 45300.0ohm Tolerance= 0.5%	1
6.	R3_S3	Generic	Ideal	Res= 459.0ohm Tolerance= 0.5%	1